Questions: Introduction to sigma notation

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Summary

Questions relating to the guide on introduction to sigma notation.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to sigma notation.

Q1

Calculate the value of the following sums in sigma notation. You may use the properties of sums but they should not be necessary.

1.1. $\sum_{i=1}^{10} 2i$

1.2. $\sum_{i=2}^{11} i$

1.3. $\sum_{i=3}^{6} 3i$

1.4. $\sum_{i=1}^{5} i^3$

1.5. $\sum_{i=2}^{6} 5i^2$

1.6. $\sum_{i=3}^{6} 2$

1.7. $\sum_{i=1}^{6} j$

Q2

Express the following using sigma notation. Note that there are multiple correct answers for some of the questions. It is recommended to use i as your variable so that your answers will align with those provided.

2.1. $3 + 6 + 9 + 12$

2.2. $-1 - 2 - 3 - 4$

2.3. $0 + 3 + 9 + 27 + 81$

2.4. $1 + 1 + 1 + 1 + 1$

2.5. $6 - 12 + 18 - 24$

2.6. $8 + 16 + 12 + 4$

2.7. $25 + 20 + 15 + 10 + 5$

Q3

Using the properties listed in the guide write the following sums in their simplest form; that is, with as little information as possible within the summation.

3.1. $\sum_{i=1}^{n} 2i$

3.2. $\sum_{i=1}^{n} 2i + \sum_{j=1}^{n} 2i$

3.3. $\sum_{i=0}^{n} 4i + \sum_{i=1}^{n} 2i$

3.4. $\sum_{i=2}^{n} 2i - \sum_{i=1}^{n} i$

After attempting the questions above, please click this link to find the answers.

Version history and licensing

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